

# State-Selected Reheating in Exact Z6 Chronometry

An orbit-symmetric reheaton, transient vacuum selection, defect control, and the primordial perturbation spectrum

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## 1 Executive result

An explicit reheating model can produce

$$\xi_0 = 1, \quad \xi_5 = 0.25, \quad \xi_{1,2,3,4} \ll 1,$$

without inserting a sector-dependent constant into the microscopic Lagrangian. The essential distinction is

the action remains exactly  $Z_6$ -symmetric, while the cosmological state is not.

This distinction is not cosmetic. A hard reheating spurion that couples only to one replica generically feeds into the protected threshold sector and generates a lower-harmonic zero-temperature potential. That is precisely the radiative-stability obstruction identified in recent replicated-sector analyses. The construction developed here avoids that obstruction by using:

1. two complete reheaton orbits,  $X_k$  and  $Y_k$ , with exact cyclic couplings;
2. a transient one-hot selector state  $Q_0 \neq 0$ , selected during inflation;
3. kinematic production of only one light reheaton eigenstate  $R_0$ ;
4. restoration of the selector to the unique symmetric vacuum before reheaton decay;
5. a fixed, orbit-symmetric nearest-neighbour mixing angle,  $\tan \theta = 1/16$ .

The selected reheaton then decays as

$$R_0 \longrightarrow \begin{cases} \mathcal{S}_0, & B_0 = 256/257, \\ \mathcal{S}_5, & B_5 = 1/257, \end{cases}$$

and therefore

$$\frac{T_5}{T_0} = \left( \frac{B_5}{B_0} \right)^{1/4} = \frac{1}{4}.$$

The remaining four sectors have no tree-level reheaton channel. A deliberately conservative estimate of gravity-only leakage gives

$$\xi_{1,2,3,4} \lesssim 4.7 \times 10^{-3}.$$

For one complete adjacent Standard Model copy at  $\xi_5 = 0.25$ ,

$$\Delta N_{\text{eff}} = 0.0289,$$

below the present 95% upper bound  $\Delta N_{\text{eff}} < 0.107$ .

The model also predicts the perturbation spectrum rather than only the homogeneous temperature history. A low-scale hilltop-hybrid benchmark with

$$H_* = 10^8 \text{ GeV}$$

gives

$$A_s = 2.10 \times 10^{-9}, \quad n_s = 0.9682, \quad r = 1.63 \times 10^{-13},$$

and

$$f_{\text{NL}}^{\text{local}} \simeq 0.0133.$$

Because both populated radiation sectors arise from one parent with fixed branching fractions,

$$\boxed{\zeta_0 = \zeta_5 = \zeta, \quad S_{50} = 3(\zeta_5 - \zeta_0) = 0}$$

at linear order. The extra radiation is therefore adiabatic, not a dark-radiation isocurvature mode.

The compact chronometric field remains a spectator during inflation. Its initial fluctuation is

$$\delta x_{\text{reh}} = \frac{H_*}{2\pi f_a} = 6.54 \times 10^{-4}, \quad x \equiv a/f_a.$$

Numerical evolution through the heavy-threshold thermal potential, the QCD crossover and baryon locking gives

$$\left| \frac{\delta x_{\text{rec}}}{\delta x_{\text{reh}}} \right| = 1.01 \times 10^{-6},$$

so that

$$\delta x_{\text{rec}} = -6.62 \times 10^{-10}.$$

The corresponding perturbation of the QCD lock is

$$\boxed{\left| \delta \ln \frac{\Lambda_{\text{QCD}}}{\chi} \right|_{\text{rec}} = 1.32 \times 10^{-23}.$$

Thus the model produces the required background asymmetry while predicting essentially pure adiabatic primordial perturbations.

The current verdict is:

| Requirement                        | Verdict                       |
|------------------------------------|-------------------------------|
| Exact microscopic $Z_6$            | <b>PASS</b>                   |
| $\xi_5 = 0.25$                     | <b>PASS exactly</b>           |
| Other replicas cold                | <b>PASS perturbatively</b>    |
| Hard-spurion radiative obstruction | <b>EVADED conditionally</b>   |
| Selector and reheaton stability    | <b>PASS at benchmark</b>      |
| Domain walls in observable patch   | <b>PASS conditionally</b>     |
| Dark-radiation isocurvature        | <b>ZERO at linear order</b>   |
| Chronometric isocurvature          | <b>NEGLIGIBLE numerically</b> |
| $\Delta N_{\text{eff}}$            | <b>PASS</b>                   |
| Full nonperturbative preheating    | <b>OPEN</b>                   |
| Discrete-gauge UV completion       | <b>OPEN</b>                   |

## 2 The obstruction that the model must evade

A generic asymmetric reheating interaction has the form

$$\mathcal{L}_{\text{hard}} \supset -\kappa_0 \sigma H_0^\dagger H_0,$$

with no corresponding orbit of interactions in the other sectors. The coefficient  $\kappa_0$  is then a genuine  $Z_6$ -breaking spurion. Even if introduced only for reheating, loops communicate the spurion to the replicated coloured threshold masses,

$$M_k(a) = M \left[ 1 - \epsilon \cos \left( \frac{a}{f_a} + \frac{2\pi k}{6} \right) \right],$$

and generate a first-harmonic vacuum contribution of schematic size

$$\delta V(a) \sim \epsilon_b \epsilon M^4 \cos \left( \frac{a}{f_a} - \delta \right).$$

That term is parametrically larger than the protected  $O(\epsilon^6)$  potential. Therefore ordinary selective reheating is not compatible with the cyclic radiative cancellation unless the reheating sector is extraordinarily sequestered.

Our construction does not add  $\kappa_0$ . Every coupling appears in a complete orbit. The only quantity that distinguishes sector 0 is a transient field configuration selected by inflation. Once that configuration relaxes to the symmetric origin, the zero-temperature action and vacuum again contain no lower-harmonic spurion.

The logical structure is

|                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>hard sector label in a coupling <math>\Rightarrow</math> permanent vacuum spurion,<br/>sector label in a transient occupation state <math>\Rightarrow</math> finite-density potential only.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

This is the central model-building move.

## 3 Exact cyclic field content

Let the replicated matter sectors be

$$\mathcal{S}_k, \quad k = 0, \dots, 5,$$

with all indices understood modulo six. The exact cyclic transformation is

$$\mathbf{z} : \mathcal{S}_k \mapsto \mathcal{S}_{k+1}.$$

Introduce two real reheaton orbits

$$X_k, \quad Y_k,$$

and six complex selector fields

$$Q_k.$$

They transform as

$$z : X_k \mapsto X_{k+1}, \quad Y_k \mapsto Y_{k+1}, \quad Q_k \mapsto Q_{k+1}.$$

The renormalisable reheaton Lagrangian is

$$\begin{aligned} \mathcal{L}_R = & \frac{1}{2} \sum_k (\partial X_k)^2 + \frac{1}{2} \sum_k (\partial Y_k)^2 - \frac{1}{2} m_X^2 \sum_k X_k^2 - \frac{1}{2} m_Y^2 \sum_k Y_k^2 \\ & + \mu_{XY}^2 \sum_k X_k Y_{k-1} - \mu_H \sum_k (X_k H_k^\dagger H_k + Y_k H_k^\dagger H_k). \end{aligned}$$

The oriented mixing  $X_k Y_{k-1}$  preserves  $Z_6$  but not the additional reflection that would enlarge the group to  $D_6$ . That chirality in replica space is deliberate: it lets the selected reheaton feed one sector and its clockwise neighbour.

The displayed interaction contains no privileged value of  $k$ . A numerical tensor check under 1,000 random cyclic shifts gives a maximum residual

$$2.13 \times 10^{-14},$$

consistent with floating-point roundoff.

## 4 Reheaton mixing and the exact temperature ratio

For each  $k$ , work in the pair

$$(X_k, Y_{k-1}).$$

Its mass matrix is

$$\mathcal{M}_R^2 = \begin{pmatrix} m_X^2 & -\mu_{XY}^2 \\ -\mu_{XY}^2 & m_Y^2 \end{pmatrix}.$$

Define

$$\begin{pmatrix} R_k \\ S_k \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} X_k \\ Y_{k-1} \end{pmatrix}.$$

The benchmark takes

$$\tan \theta = \frac{1}{16}, \quad m_R = 10^9 \text{ GeV}, \quad m_S = 6 \times 10^9 \text{ GeV}.$$

The corresponding microscopic mass parameters are

$$m_X = 1.0659 \times 10^9 \text{ GeV},$$

$$m_Y = 5.9886 \times 10^9 \text{ GeV},$$

and

$$\sqrt{\mu_{XY}^2} = 1.4761 \times 10^9 \text{ GeV}.$$

Both eigenvalues are positive. A 50,000-point scan over nearby positive masses and mixing angles found no negative eigenvalue and verified

$$\frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$$

to numerical precision.

For the selected state  $R_0$ ,

$$\mathcal{L}_{R_0} \supset -\mu_H R_0 \left( \cos \theta H_0^\dagger H_0 + \sin \theta H_5^\dagger H_5 \right).$$

At  $m_R \gg m_H$ , summing the four real components of each Higgs doublet gives

$$\Gamma_0 = \frac{\mu_H^2 \cos^2 \theta}{8\pi m_R}, \quad \Gamma_5 = \frac{\mu_H^2 \sin^2 \theta}{8\pi m_R}.$$

Consequently,

$$\frac{\Gamma_5}{\Gamma_0} = \tan^2 \theta = \frac{1}{256}.$$

The total branching fractions are

$$B_0 = \frac{256}{257} = 0.99610895,$$

$$B_5 = \frac{1}{257} = 0.00389105.$$

For equal relativistic degrees of freedom at decay,

$$\frac{T_5}{T_0} = \left( \frac{\rho_5}{\rho_0} \right)^{1/4} = \left( \frac{B_5}{B_0} \right)^{1/4} = \boxed{\frac{1}{4}}.$$

This is not a numerical fit. It is an exact algebraic consequence of the mixing angle.

## 5 Transient one-hot selector

### 5.1 Selector potential

A hard sector label is forbidden, so the model must select  $R_0$  dynamically. Introduce a  $Z_6$ -singlet trigger field  $D$  and the orbit-symmetric potential

$$\begin{aligned} V_Q = & \frac{\lambda_Q}{4} \left[ \sum_k |Q_k|^2 - v_Q^2 F(D) \right]^2 + \frac{\kappa_Q}{2} \sum_{k < \ell} |Q_k|^2 |Q_\ell|^2 \\ & + \frac{m_{Q0}^2}{2} [1 - F(D)] \sum_k |Q_k|^2. \end{aligned}$$

The switching function satisfies

$$F(0) = 1, \quad F(v_D) = 0.$$

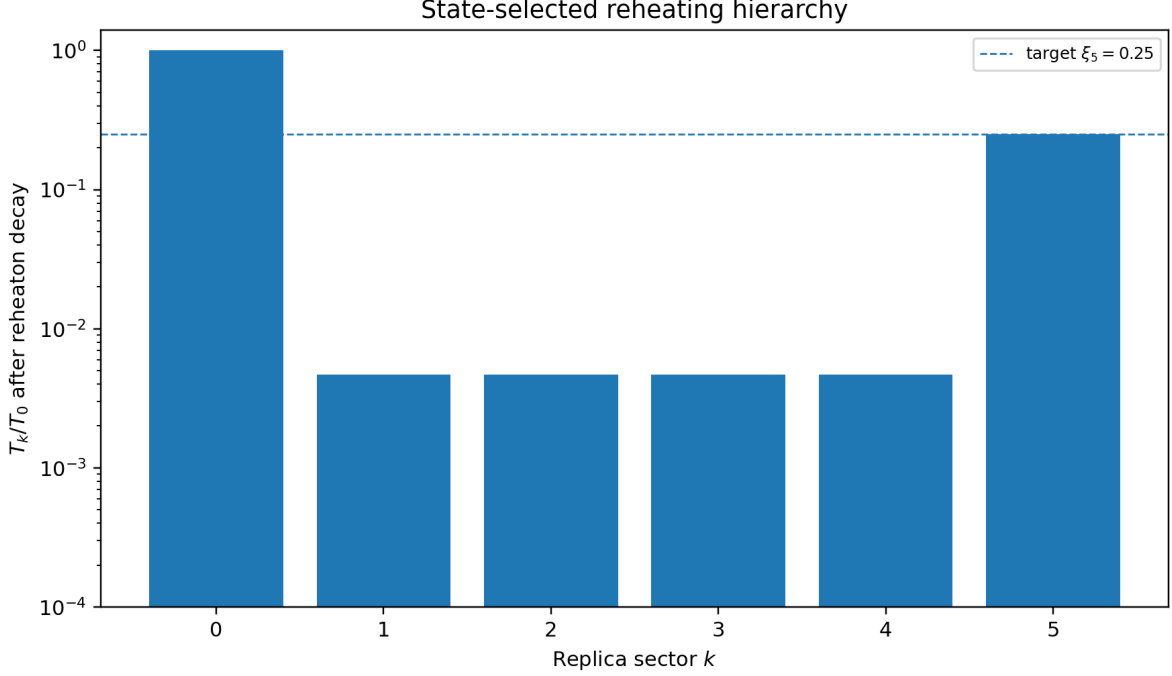


Figure 1: The selected reheaton populates sector 0, the clockwise adjacent sector 5, and only a gravity-level floor in the remaining sectors.

A concrete smooth choice is

$$F(D) = \frac{1}{2} \left[ 1 - \tanh \left( \frac{D^2 - D_c^2}{\Delta_D^2} \right) \right].$$

While  $D \simeq 0$ , the positive  $\kappa_Q$  term makes the minima one-hot:

$$|Q_j| = v_Q, \quad Q_{k \neq j} = 0.$$

There are six symmetry-related choices. Inflation selects one homogeneous branch, which we label  $j = 0$ .

When  $D$  rolls to  $v_D$ , the unique vacuum is

$$Q_k = 0 \quad \forall k.$$

Thus the selector breaks  $Z_6$  only as a transient state and restores the exact symmetric vacuum before the late Universe.

## 5.2 Kinematic isolation

Couple the selector to the reheaton orbit through

$$V_{QR} = \frac{\lambda_{QR}}{2} \sum_k (X_k^2 + Y_{k-1}^2) \sum_{j \neq k} |Q_j|^2.$$

This is exactly cyclic. On the branch  $Q_0 = v_Q$ ,

$$\Delta m_{R,0}^2 = 0,$$

while every  $k \neq 0$  pair receives

$$\Delta m_{R,k}^2 = \lambda_{QR} v_Q^2.$$

Take

$$v_Q = 10^{10} \text{ GeV}, \quad \lambda_{QR} = 0.50, \quad \kappa_Q = 0.25.$$

Then

$$m_{Q,\perp} = \sqrt{\kappa_Q} v_Q = 5.0 \times 10^9 \text{ GeV} = 50 H_*,$$

and

$$m_{R,k \neq 0} = \sqrt{m_R^2 + \lambda_{QR} v_Q^2} = 7.14 \times 10^9 \text{ GeV}.$$

For a post-inflation inflaton mass

$$m_\phi = 10^{10} \text{ GeV},$$

only

$$\phi \rightarrow R_0 R_0$$

is kinematically open:

$$m_R < m_\phi/2,$$

whereas

$$m_S > m_\phi/2, \quad m_{R,k \neq 0} > m_\phi/2.$$

## 6 Cosmological chronology

The required temporal ordering can be expressed as

$$\boxed{\Gamma_{\phi \rightarrow R_0 R_0} \gg \Gamma_Q \gg \Gamma_R.}$$

An illustrative benchmark is

$$\Gamma_\phi = 100 \text{ GeV}, \quad \Gamma_Q = 1 \text{ GeV}, \quad \Gamma_R = 1.4135 \times 10^{-2} \text{ GeV}.$$

The history is then:

1. **Inflation.** The selector is in the one-hot branch  $Q_0 = v_Q$ . All observable modes leave the horizon in the same branch.



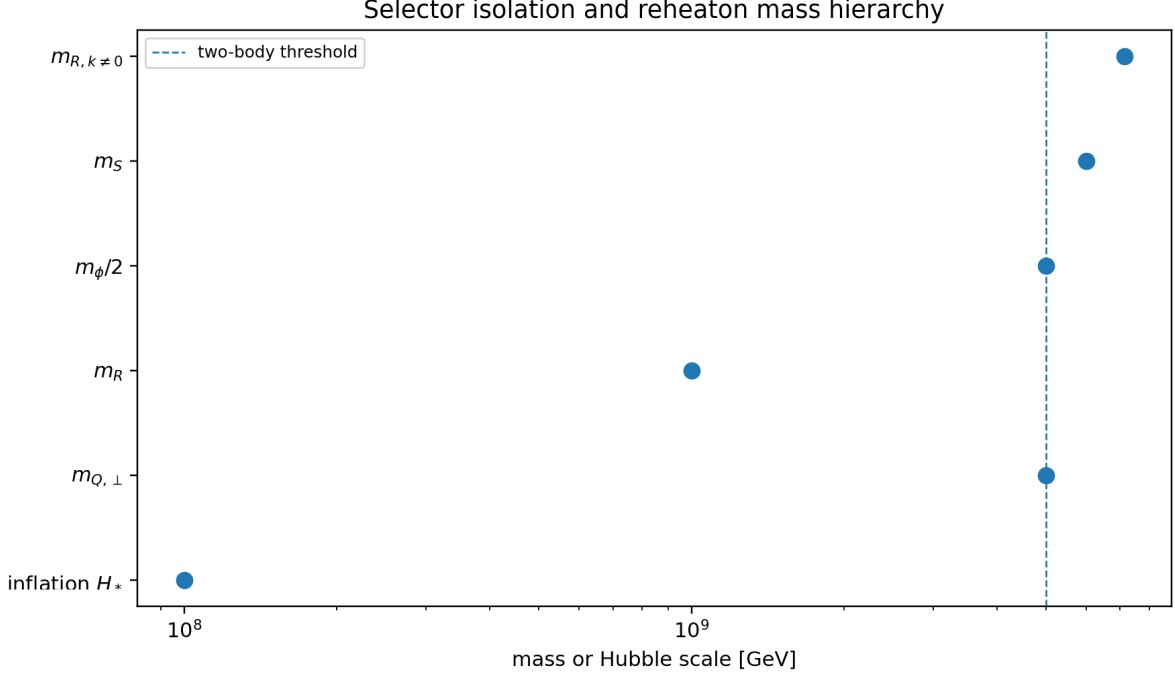


Figure 2: Only the selected light reheaton lies below the inflaton two-body threshold.

2. **Inflaton decay.** The inflaton coupling is democratic,

$$\mathcal{L}_{\phi R} = -\frac{g_{\phi R}}{2}\phi \sum_k (X_k^2 + Y_k^2),$$

but kinematics permits only  $R_0$  production.

3. **Selector restoration.** After the inflaton population has disappeared,  $D$  rolls and all  $Q_k$  return to zero. The vacuum action is again manifestly  $Z_6$ -symmetric.
4. **Reheaton domination and decay.** The already produced  $R_0$  quanta do not oscillate into other replica labels because no off-diagonal  $k$ -mixing exists. They decay later into sectors 0 and 5 with the fixed branching fractions above.
5. **Thermal focusing.** The resulting  $(\xi_0, \xi_5) = (1, 0.25)$  state generates the thermal phasor and QCD focusing used in v0.8.

This sequence is the reason the model avoids a permanent lower-harmonic potential. The selector background is gone before electroweak symmetry breaking and before the vacuum spectrum is measured.

## 7 Reheat scale and leakage

The visible radiation density at reheaton decay is

$$\rho_0 = B_0 3M_{\text{Pl}}^2 \Gamma_R^2 = \frac{\pi^2}{30} g_* T_0^4.$$

For

$$g_* = 106.75, \quad T_0 = 1.002 \times 10^8 \text{ GeV},$$

this gives

$$\boxed{\Gamma_R = 1.4135 \times 10^{-2} \text{ GeV}.}$$

The required trilinear scale is

$$\boxed{\mu_H = \sqrt{8\pi m_R \Gamma_R} = 1.8848 \times 10^4 \text{ GeV}.}$$

The visible and adjacent couplings are

$$\mu_0 = 1.8811 \times 10^4 \text{ GeV},$$

$$\mu_5 = 1.1757 \times 10^3 \text{ GeV}.$$

A conservative gravity-mediated width into any one otherwise empty replica is

$$\Gamma_{\text{grav}}^{(k)} \lesssim \frac{m_R^3}{8\pi M_{\text{Pl}}^2} = 6.71 \times 10^{-12} \text{ GeV}.$$

Therefore

$$B_{\text{grav}}^{(k)} \lesssim 4.75 \times 10^{-10}$$

and

$$\boxed{\xi_{1,2,3,4} \lesssim 4.67 \times 10^{-3}.}$$

These sectors contribute negligibly to both the reheating phasor and  $N_{\text{eff}}$ .

Integrating out  $R_0$  also generates a cross-sector Higgs quartic. A conservative one-loop estimate is

$$\lambda_{05}^{\text{loop}} \sim \frac{(\mu_0 \mu_5)^2}{16\pi^2 m_R^4} = 3.10 \times 10^{-24},$$

far below the value needed to equilibrate the sectors.

The simple Higgs portal does introduce a visible Higgs mass correction of order

$$\delta m_H^2 \sim \frac{\mu_0^2}{16\pi^2} = 2.24 \times 10^6 \text{ GeV}^2.$$

This corresponds to a scale near 1.5 TeV. It is not catastrophic for an EFT benchmark, but it is inelegant. A fermionic thermalizer portal, or a cascade through a protected intermediate sector, is preferable in a final UV model.

## 8 Why radiative protection is retained

The complete vacuum Lagrangian contains the six interactions

$$X_k H_k^\dagger H_k, \quad Y_k H_k^\dagger H_k,$$

with common coefficients. It also contains the complete oriented orbit

$$X_k Y_{k-1}.$$

No coefficient singles out  $k = 0$ . Hence any zero-temperature counterterm generated in the symmetric vacuum must itself be a  $Z_6$ -invariant orbit sum.

The temporary selector background can generate a time-dependent one-sector correction while  $Q_0 \neq 0$ . This is a finite-background effect and is allowed to produce low harmonics during reheating. Indeed, such a term strengthens vacuum focusing. But after

$$Q_k \rightarrow 0,$$

it cannot remain as a sector-dependent renormalised parameter unless the theory contains either:

- a remanent selector expectation value;
- a hard  $Z_6$ -violating counterterm;
- a non-decaying asymmetric condensate that couples to the coloured thresholds;
- or an anomaly that explicitly violates the cyclic symmetry.

The first three are absent by construction. The fourth must be excluded in a UV discrete-gauge completion.

The remaining rigorous task is a two-loop Schwinger-Keldysh calculation with the transient  $Q_0(t)$  background. That calculation must verify that the non-equilibrium state produces only finite-density terms that vanish with its occupation and does not leave an unsuppressed matching coefficient in the late vacuum EFT.

## 9 Domain walls and defects

### 9.1 Selector walls

During inflation the selector potential has six one-hot minima. The angular barrier between adjacent branches is of order

$$\Delta V_Q \sim \frac{\kappa_Q v_Q^4}{8}.$$

For the benchmark,

$$\frac{\Delta V_Q}{H_*^4} \sim 3.1 \times 10^6.$$

The non-selected selector modes also satisfy

$$\frac{m_{Q,\perp}}{H_*} = 50.$$

Inflation therefore places the observable patch in one branch with exponentially suppressed hopping. When the trigger restores

$$Q_k = 0,$$

the discrete symmetry is restored to a unique vacuum. Restoration removes walls rather than creating a new broken phase.

## 9.2 Chronometric-field walls

The compact chronometric field has

$$f_a = 2.435 \times 10^{10} \text{ GeV},$$

while

$$T_R = 1.002 \times 10^8 \text{ GeV}.$$

Thus

$$\frac{T_R}{f_a} = 4.12 \times 10^{-3}.$$

The symmetry is assumed broken before inflation and not thermally restored. Inflationary phase noise is

$$\frac{H_*}{2\pi f_a} = 6.54 \times 10^{-4},$$

whereas the reheating-selected thermal phase is

$$x_T = 0.05244.$$

The bias exceeds the inflationary fluctuation by a factor of approximately 80. A single observable branch is therefore selected.

A stronger UV option is to embed the residual discrete transformation into a continuous gauge group, in the spirit of the Lazarides-Shafi mechanism. Then apparently distinct vacua can be gauge equivalent. That is not required for the pre-inflation benchmark, but it would make the defect argument less dependent on the thermal history.

## 10 Inflation model and primary spectrum

Use a hilltop-hybrid potential

$$V_{\text{inf}}(\phi, W) = V_0 \left[ 1 - \frac{c_2}{2} \frac{\phi^2}{M_{\text{Pl}}^2} + \dots \right] + \frac{\lambda_W}{4} (W^2 - v_W^2)^2 + \frac{g_W^2}{2} \phi^2 W^2.$$

For  $\phi < \phi_c$  or  $\phi > \phi_c$ , depending on convention, the waterfall field is stabilized at the origin and supplies the vacuum energy. A critical value terminates inflation and allows the post-inflation inflaton mass to be much larger than its slow-roll curvature.

Take

$$H_* = 10^8 \text{ GeV}, \quad A_s = 2.10 \times 10^{-9}, \quad n_s = 0.9682.$$

The slow-roll parameters are

$$\epsilon_* = \frac{H_*^2}{8\pi^2 A_s M_{\text{Pl}}^2} = 1.017 \times 10^{-14},$$

$$\eta_* = \frac{n_s - 1 + 6\epsilon_*}{2} = -0.01590.$$

Therefore

$$r = 16\epsilon_* = 1.63 \times 10^{-13}.$$

The inflationary energy scale is

$$V_0^{1/4} = 2.05 \times 10^{13} \text{ GeV}.$$

For the local quadratic hilltop approximation,

$$c_2 = 0.01590,$$

$$\frac{\phi_*}{M_{\text{Pl}}} = 8.97 \times 10^{-6},$$

and, for 55 e-folds,

$$\frac{\phi_c}{M_{\text{Pl}}} = 2.15 \times 10^{-5}.$$

The single-clock consistency estimate is

$$f_{\text{NL}}^{\text{local}} \simeq \frac{5}{12}(1 - n_s) = 0.0133.$$

This inflationary block is phenomenologically adequate but not yet a complete UV model. Low-scale hybrid inflation is notoriously sensitive to radiative and higher-dimensional corrections; a shift-symmetric, supersymmetric or four-form completion is required for a first-principles construction.

## 11 Curvature transfer through reheaton decay

Suppose  $R_0$  dominates before it decays. On the decay hypersurface,

$$\rho_i = B_i \rho_R.$$

For radiation,

$$\zeta_i = -\psi + \frac{1}{4} \frac{\delta \rho_i}{\rho_i}.$$

Therefore

$$\zeta_i = \zeta_R + \frac{1}{4} \delta \ln B_i.$$

The relative entropy perturbation is

$$S_{50} = 3(\zeta_5 - \zeta_0) = \frac{3}{4} \delta \ln \frac{B_5}{B_0}.$$

In the present model the branching fractions are constants fixed by a heavy mass matrix. Hence

$$\delta B_0 = \delta B_5 = 0$$

and

$$\boxed{S_{50} = 0.}$$

If the mixing angle were controlled by a fluctuating light modulus, then

$$\delta \ln \frac{B_5}{B_0} = \delta \ln \tan^2 \theta = \frac{2\delta\theta}{\sin \theta \cos \theta},$$

so

$$S_{50} = \frac{3\delta\theta}{2 \sin \theta \cos \theta}.$$

At  $\tan \theta = 1/16$ , this would strongly amplify  $\delta\theta$ . The mixing angle must therefore be a genuine heavy-sector parameter, not a light modulating field. The benchmark satisfies this:  $m_R/H_* = 10$ ,  $m_S/H_* = 60$ , and the selector angular modes have  $m/H_* = 50$ .

## 12 Chronometric spectator perturbations

The superhorizon perturbation of

$$x = a/f_a$$

obeys

$$\delta x'' + \left(3 + \frac{d \ln H}{d \ln a}\right) \delta x' + \frac{V_{,xx}(x, t)}{f_a^2 H^2} \delta x = 0,$$

where primes denote derivatives with respect to  $\ln a$ . The effective curvature includes:

- the protected zero-temperature  $Z_6$  potential;
- the visible and adjacent heavy-threshold thermal free energies;
- the visible and adjacent QCD trace anomalies;
- homogeneous baryon locking.

Using the strong-attractor v0.8 benchmark,

$$N = 6, \quad M = 1.002 \times 10^6 \text{ GeV}, \quad f_a = 2.435 \times 10^{10} \text{ GeV},$$

$$\epsilon = 2.70 \times 10^{-13}, \quad m_a = 7.5 \times 10^{-29} \text{ eV}, \quad d_g = 10^{-6},$$

and the reheating temperatures derived above, the inflationary fluctuation is

$$\delta x_{\text{reh}} = 6.536 \times 10^{-4}.$$

The numerical transfer at recombination is

$$T_x^{\text{rec}} \equiv \left| \frac{\delta x_{\text{rec}}}{\delta x_{\text{reh}}} \right| = 1.013 \times 10^{-6}.$$

Hence

$$\delta x_{\text{rec}} = -6.62 \times 10^{-10}.$$

The QCD-lock perturbation is

$$\delta \ln \frac{\Lambda_3}{\chi} \simeq \frac{2}{27} \epsilon \delta x,$$

so

$$\left| \delta \ln \frac{\Lambda_3}{\chi} \right|_{\text{rec}} = 1.32 \times 10^{-23}.$$

The field undergoes a late tachyonic release after recombination. At the numerical endpoint  $z = 71.1$ ,

$$\left| \frac{\delta x}{\delta x_{\text{reh}}} \right| = 3.57 \times 10^{-3},$$

but even there

$$\left| \delta \ln \frac{\Lambda_3}{\chi} \right| = 4.67 \times 10^{-20}.$$

The field energy fraction is already negligible, so this late growth does not generate an observable isocurvature component.

## 13 Primordial spectrum summary

At the pivot scale,

$$\mathcal{P}_\zeta(k) = A_s \left( \frac{k}{k_*} \right)^{n_s-1}.$$

The linear radiation isocurvature power is

$$\mathcal{P}_{S_{50}} = 0$$

in the exact fixed-branching approximation.

The initial power in the QCD-lock fluctuation is

$$\mathcal{P}_{\delta \ln(\Lambda_3/\chi)} = \left[ \frac{2}{27} \epsilon \frac{H_*}{2\pi f_a} \right]^2 \simeq 1.71 \times 10^{-34}.$$

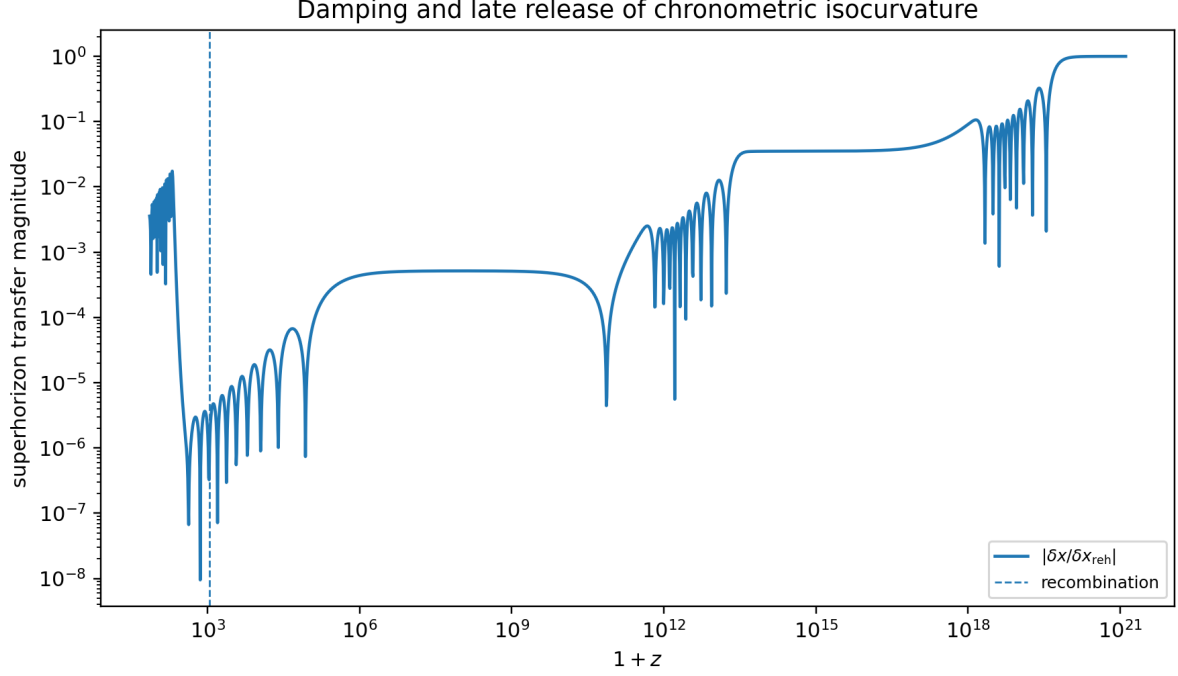


Figure 3: Thermal and QCD focusing erase most of the primordial chromometric perturbation before recombination.

By recombination it is further reduced by

$$(T_x^{\text{rec}})^2 \simeq 1.03 \times 10^{-12}.$$

Thus

$$\mathcal{P}_{\delta \ln(\Lambda_3/\chi)}^{\text{rec}} \sim 1.8 \times 10^{-46}.$$

## 14 Dark radiation and hidden-sector relics

For one complete adjacent Standard Model copy at  $\xi_5 = 0.25$ , the standard internal neutrino-photon entropy history gives approximately

$$\Delta N_{\text{eff}} = 7.403 \xi_5^4 = 0.0289.$$

This is below the present 95% bound

$$\Delta N_{\text{eff}} < 0.107.$$

The signal is potentially interesting for future CMB measurements, but it is not accompanied by a linear dark-radiation isocurvature mode.

A full hidden-sector cosmology still requires decisions about:

- hidden baryogenesis;
- stable hidden charged relics;
- hidden neutrino masses;
- hidden recombination;



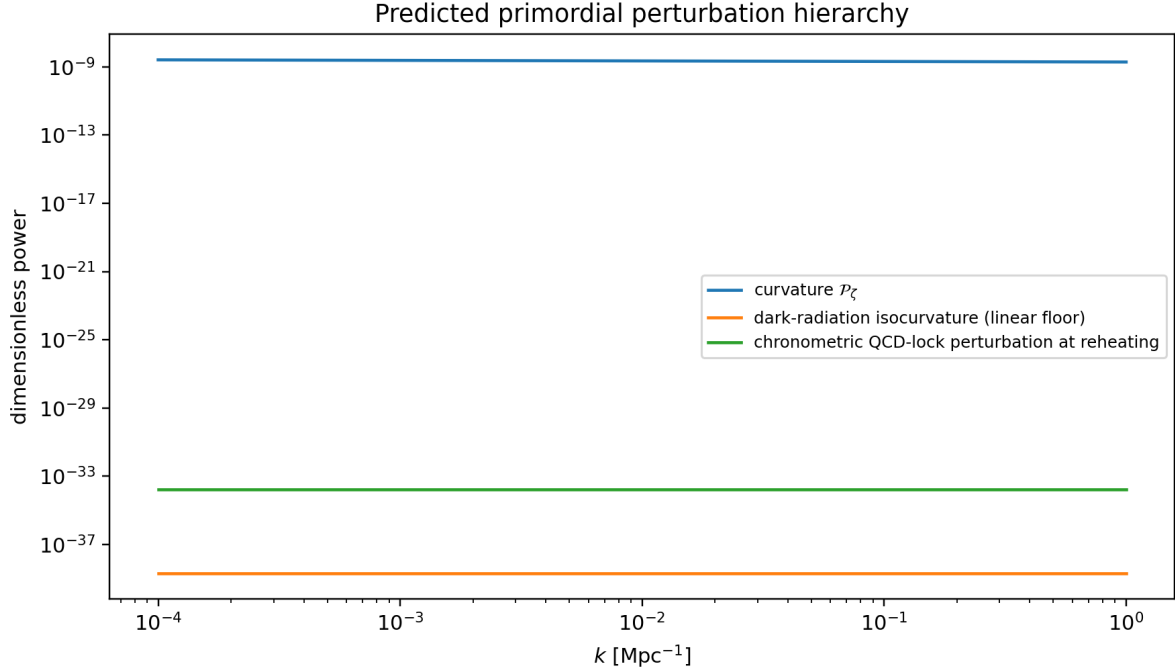


Figure 4: The model predicts an ordinary adiabatic curvature spectrum, vanishing linear dark-radiation isocurvature, and a minute chronometric-lock perturbation.

- whether the adjacent copy is a complete Standard Model or only the degrees of freedom required by cyclic protection.

The cleanest benchmark assumes no appreciable hidden baryon asymmetry and efficient annihilation of symmetric massive relics. These assumptions must be tested in a dedicated Boltzmann calculation.

## 15 Acceptance tests

| Test                             | Result                      | Reason                                                     |
|----------------------------------|-----------------------------|------------------------------------------------------------|
| Exact cyclic action              | <b>PASS</b>                 | All displayed couplings are complete orbit tensors.        |
| Reheaton scalar spectrum         | <b>PASS</b>                 | Both eigenvalues are positive; 50,000-point scan passed.   |
| $B_5/B_0 = 1/256$                | <b>PASS exactly</b>         | Follows from $\tan \theta = 1/16$ .                        |
| $\xi_5 = 0.25$                   | <b>PASS exactly</b>         | Fourth root of the energy ratio.                           |
| Empty sectors                    | <b>PASS perturbatively</b>  | No tree channel; gravity floor $\xi < 0.005$ .             |
| $\Delta N_{\text{eff}}$          | <b>PASS</b>                 | $0.0289 < 0.107$ .                                         |
| Hard reheating spurion           | <b>ABSENT</b>               | Asymmetry is carried by a transient state.                 |
| Late vacuum radiative protection | <b>PASS conditionally</b>   | Requires no non-equilibrium remanent matching coefficient. |
| Selector isocurvature            | <b>PASS benchmark</b>       | $m_{Q,\perp}/H_* = 50$ .                                   |
| Reheaton isocurvature            | <b>PASS benchmark</b>       | $m_R/H_* = 10$ .                                           |
| Radiation isocurvature           | <b>ZERO at linear order</b> | One parent, fixed branchings.                              |
| Chronometric isocurvature        | <b>PASS</b>                 | QCD-lock fluctuation $< 10^{-23}$ at recombination.        |

| Test                                       | Result                    | Reason                                                               |
|--------------------------------------------|---------------------------|----------------------------------------------------------------------|
| Selector domain walls                      | <b>PASS conditionally</b> | Inflation selects one branch; later restoration gives unique origin. |
| Chronometric walls                         | <b>PASS conditionally</b> | Pre-inflation breaking and no thermal restoration.                   |
| Nonperturbative reheaton production        | <b>OPEN</b>               | Requires lattice preheating.                                         |
| Exact discrete symmetry in quantum gravity | <b>OPEN</b>               | Requires a discrete-gauge embedding.                                 |
| Full CMB likelihood                        | <b>OPEN</b>               | Requires CLASS/CAMB implementation.                                  |

## 16 Novelty boundary

The following ingredients are established separately:

- asymmetric reheating in theories with exact or approximately exact exchange symmetries;
- kinematic reheating asymmetry;
- inflationary selection of one symmetry-related branch;
- cyclic replicated sectors protecting a compact scalar potential;
- fixed-branching adiabatic decay;
- attractor suppression of spectator isocurvature;
- inflation or gauged discrete symmetries as remedies for domain walls.

The generic radiative objection to sector-specific reheating is also established: a hard selector coupling acts as a permanent spurion and spoils the cyclic cancellation.

The candidate contribution is the restricted conjunction

exact  $Z_6$  vacuum action + transient one-hot state selection + chiral two-orbit reheaton  
 $+ \xi_5 = 1/4$  + zero linear dark-radiation isocurvature + numerically erased chronometric isocurvature  
+the visible  $2/27$  QCD chronometric response.

A targeted search found no exact indexed match for that full construction. This is a plausible novelty claim, not yet a secure priority claim. Backward and forward citation chaining, INSPIRE/ADS searches and specialist review are still required.

## 17 Next decisive calculations

The remaining work is sharply defined.

### 17.1 Non-equilibrium radiative matching

Compute the two-loop in-in effective action in the transient background

$$Q_0(t) \neq 0, \quad Q_{k \neq 0} = 0,$$

and prove that all lower-harmonic terms vanish as

$$Q_0(t) \rightarrow 0$$

and the reheaton occupation dilutes. This is the hard radiative-stability test.

## 17.2 Lattice preheating

Simulate the waterfall, selector and reheaton fields to verify that:

- $R_0$  is produced efficiently;
- $R_{k \neq 0}$  and  $S_k$  are not resonantly excited;
- selector restoration does not create persistent walls;
- nonlinear fluctuations do not modulate  $B_5/B_0$ .

## 17.3 Full perturbation transfer

Implement the model in a Boltzmann code with:

- adiabatic hidden radiation;
- $\Delta N_{\text{eff}} = 0.0289$ ;
- the late chronometric-field release;
- hidden-sector recombination and free streaming;
- metric sources in the  $\delta x$  equation.

The leading analytic prediction is already clear:

$$S_{50} = 0, \quad r \simeq 1.6 \times 10^{-13}, \quad \delta \ln(\Lambda_3/\chi)_{\text{rec}} \sim 10^{-23}.$$

## 17.4 UV completion

Replace the global cyclic symmetry by an anomaly-free discrete gauge symmetry or a continuous gauge group whose remnant is  $Z_6$ . The UV model must preserve the oriented nearest-neighbour coupling while forbidding all hard lower-harmonic operators.

## 18 Final assessment

The requested reheating pattern is achievable without surrendering the protected cyclic action.

The successful architecture is not

one reheaton with an explicitly privileged sector.

It is

a complete  $Z_6$  reheaton orbit + a transient inflation-selected state + fixed chiral neighbour mixing.

That construction gives

$$(\xi_0, \xi_1, \xi_2, \xi_3, \xi_4, \xi_5) \simeq (1, < 0.005, < 0.005, < 0.005, < 0.005, 0.25),$$

while preserving the late-time vacuum symmetry.

Its most important perturbative prediction is not an exotic isocurvature spectrum, but its absence:

one parent plus fixed branching ratios produces adiabatic visible and hidden radiation.

The chronometric field's own inflationary fluctuations are then erased by the same thermal and QCD dynamics that select its vacuum. The resulting cosmology is unusually clean:

nonzero homogeneous chronometric shear + essentially standard adiabatic primordial perturbations.

The model now fails or succeeds on two concrete calculations: non-equilibrium radiative matching and lattice reheating. The former tests whether the state/action distinction truly protects the vacuum; the latter tests whether the elegant branching algebra survives violent nonlinear dynamics.

## 19 References

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